ALGEBRA 2 · UNIT 5: RADICALS & RATIONAL EXPONENTS

# Crash Scene Investigator

*Use radical formulas and rational exponents to reconstruct an accident and study motion.*

**Honors**

2–3 class periods

Name: _____ Date: _____ Period: _____

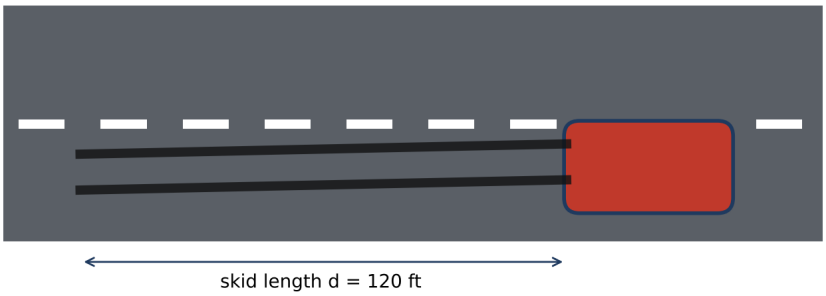
## The Scenario

A car skidded 120 feet on a dry road before stopping in a 45-mph zone. Accident investigators use s = √(30df), where s is speed in mph, d is skid length in feet, and f is the road's friction factor (dry asphalt 0.75, wet asphalt 0.45).

Your lab will estimate speeds, solve for unknowns, and explore other formulas with radicals and rational exponents, then write a report.

**Crash Lab Formula**

speed $s = \sqrt{(30 \cdot d \cdot f)}$ (mph; d in feet; f = road friction factor)




## Learning Goals

- I can use properties of rational exponents to rewrite and solve.
- I can solve equations with two radicals.
- I can analyze how measurement error affects a radical model.

## Project Timeline

| When | What you will do |
|---|---|
| Day 1 | Evidence Analysis |
| Day 2 | Harder Equations |
| Day 3 | Orbits & Report |
| Final | Investigation Report + reflection |

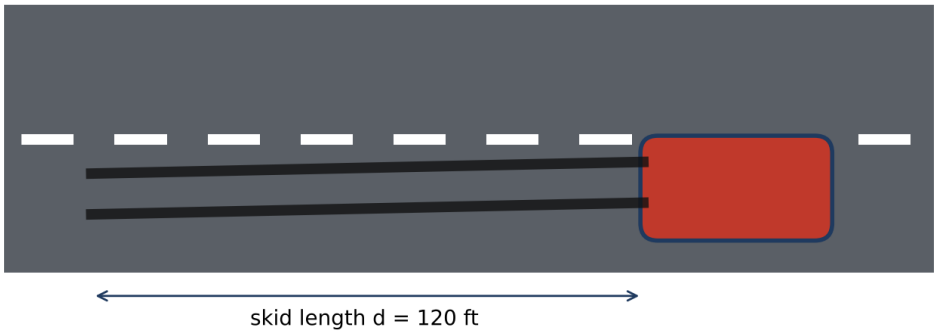
## Part 1: Evidence Analysis

**Day 1 ·** Use $s = \sqrt{(30df)}$.

**1.** If the skid length is measured 10% too short, by what percent is the speed underestimated? Use exponent properties.

**Crash Lab Formula**

speed $s = \sqrt{(30 \cdot d \cdot f)}$ (mph; d in feet; f = road friction factor)




**2.** The driver claims 45 mph. On a road with an unknown friction factor, what f would make the 120-ft skid match the claim?

**3.** Write s as a power function of d and describe how the speed grows.

## Part 2: Harder Equations

**Day 2 ·** Check every solution.

**4.** Solve $\sqrt{(x + 6)} - \sqrt{(x - 1)} = 1$.

**5.** Solve x to the ⅔ power = 9 (that is, the cube root of $x^2$, squared form) for real x.

**6.** Solve $\sqrt{(2x + 1)} = x - 1$ and identify extraneous solutions.

## Part 3: Orbits & Report

**Day 3 ·** $T = a^{1.5}$ (years, AU).

**7.** Jupiter's orbit takes 11.86 years. Find a.

**8.** Show that doubling a multiplies T by about 2.83. Write the exact factor.

**9.** Write the full report, including the error analysis.

## Final Product: Investigation Report

**Your final product must include:**

- Error analysis
- Solutions to harder radical equations
- Orbit calculations with exact factors
- Report

## Reflection

**R1.** How do rational exponents make error analysis easier?

**R2.** Why do equations with two radicals often need two squaring steps?

## Scoring Rubric

| Criteria | 4 – Exceeds | 3 – Meets | 2 – Approaching | 1 – Beginning |
|---|---|---|---|---|
| **Mathematical Accuracy** | All calculations and solutions are correct; answers include units and are checked. | Nearly all work is correct; minor errors do not change conclusions. | Several errors; some conclusions are affected. | Many errors or missing work. |
| **Reasoning & Justification** | Clear, complete explanations connect every answer to the situation; uses precise vocabulary. | Explanations are mostly clear and connected to the situation. | Explanations are brief or partly unclear. | Little or no explanation. |
| **Application & Design** | Conclusions are supported by correct radical calculations, and any extraneous solutions are identified. Goes beyond the minimum with a creative, realistic choice. | Conclusions are supported by correct radical calculations, and any extraneous solutions are identified. | Design is incomplete or only partly uses the math from the unit. | Design is missing or unrelated to the unit math. |
| **Communication & Presentation** | Report includes speed estimates with simplified radicals, solved formulas, a graph, and a conclusion about the case. Polished, organized, and visually engaging. | Report includes speed estimates with simplified radicals, solved formulas, a graph, and a conclusion about the case. Organized and easy to follow. | Product is missing required parts or is hard to follow. | Product is mostly missing. |